

## Module 2 Lesson 2

### Algebraic Expressions and Formulas (6.1)

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## Objectives

In this lesson, we will learn how to

- evaluate algebraic expressions;
- use mathematical models;
- understand the vocabulary of algebraic expressions;
- simplify algebraic expressions.

## 1. Algebraic Expressions

Algebra uses letters, called **variables**, to represent numbers.

Common variables include

$x$ ,  $y$ ,  $a$ ,  $b$ .

An **algebraic expression** is a combination of variables and numbers using operations such as addition, subtraction, multiplication, division, powers, or roots.

### Algebraic Expression

An algebraic expression may contain:

- numbers;
- variables;
- operation symbols;
- exponents or roots.

Examples of algebraic expressions include

$$3x + 5,$$

$$7x - 9y - 3,$$

$$5(x - 4)^3,$$

and

$$\frac{x + 2}{y - 1}.$$

## 2. Order of Operations Agreement

When evaluating algebraic expressions, we follow the order of operations.

### Order of Operations

1. Perform operations within the innermost parentheses and work outward.
2. If the algebraic expression involves a fraction, treat the numerator and the denominator as if they were each enclosed in parentheses.
3. Evaluate all exponential expressions.
4. Perform multiplications and divisions as they occur, working from left to right.
5. Perform additions and subtractions as they occur, working from left to right.

Example 1: Evaluate

$$7 + 5(x - 4)^3$$

for

$$x = 6.$$

Solution

Substitute 6 for  $x$ :

$$7 + 5(x - 4)^3 = 7 + 5(6 - 4)^3.$$

First simplify inside the parentheses:

$$= 7 + 5(2)^3.$$

Evaluate the exponent:

$$= 7 + 5(8).$$

Multiply:

$$= 7 + 40.$$

Add:

$$= 47.$$

### 3. Formulas and Mathematical Models

An **equation** is formed when an equal sign is placed between two algebraic expressions.

For example,

$$y = 3x + 5$$

is an equation.

A formula is an equation that uses letters to express a relationship between two or more variables.

For example, the formula for the area of a rectangle is

$$A = lw,$$

where  $A$  is area,  $l$  is length, and  $w$  is width.

#### Mathematical Modeling

Mathematical modeling is the process of finding formulas to describe real-world situations.

## 4. Modeling Caloric Needs

Mathematical models can be used to estimate quantities in real-world situations.

For example, a model may estimate the number of calories needed per day for different age groups and activity levels.

Example 2: The mathematical model

$$F = -66x^2 + 526x + 1030$$

describes the number of calories needed per day,  $F$ , by females in age group  $x$  with moderately active lifestyles.

According to the model, how many calories per day are needed by females between the ages of 19 and 30, inclusive?

The 19-30 age range is designated as group 4, so use

$$x = 4.$$

Does this underestimate or overestimate the number shown by the graph? By how much?

### Solution

Substitute 4 for  $x$ :

$$F = -66(4)^2 + 526(4) + 1030.$$

Evaluate the exponent:

$$F = -66(16) + 526(4) + 1030.$$

Multiply:

$$F = -1056 + 2104 + 1030.$$

Add:

$$F = 2078.$$

The model indicates that 2078 calories are needed per day.

## 5. Vocabulary of Algebraic Expressions

Algebraic expressions have special vocabulary.

Consider the expression

$$7x - 9y - 3.$$

### Terms

**Terms** are the parts of an algebraic expression separated by addition or subtraction.

In

$$7x - 9y - 3,$$

the terms are

$$7x, \quad -9y, \quad -3.$$

### Coefficient

A **coefficient** is the numerical part of a term.

In

$$7x - 9y - 3,$$

the coefficients of the variable terms are

$$7$$

and

$$-9.$$

### Constant

A **constant** is a term that consists of just a number.

In

$$7x - 9y - 3,$$

the constant term is

$$-3.$$

### Like Terms

**Like terms** are terms that have exactly the same variable factors.

For example,

$$7x$$

and

$$3x$$

are like terms.

### Factors

Factors are parts of each term that are multiplied.

For example, in the term

$$7x,$$

the factors are

$$7$$

and

$$x.$$

## 6. Properties of Real Numbers

Properties of real numbers help us simplify algebraic expressions.

### Commutative Property

Changing the order of addition or multiplication does not change the result.

$$a + b = b + a$$

$$ab = ba$$

### Associative Property

Changing the grouping of addition or multiplication does not change the result.

$$(a + b) + c = a + (b + c)$$

$$(ab)c = a(bc)$$

### Distributive Property

Multiplication distributes over addition or subtraction.

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

## 7. Simplifying Algebraic Expressions

To simplify an algebraic expression, we often use the distributive property and then combine like terms.

### Combining Like Terms

Only like terms can be combined.

For example,

$$5x + 3x = 8x.$$

But

$$5x + 3y$$

cannot be combined because  $x$  and  $y$  are different variables.

Example 3: Simplify:

$$5(3x - 7) - 6x.$$

### Solution

Use the distributive property:

$$5(3x - 7) - 6x = 5 \cdot 3x - 5 \cdot 7 - 6x.$$

Multiply:

$$= 15x - 35 - 6x.$$

Group like terms:

$$= (15x - 6x) - 35.$$

Combine like terms:

$$= 9x - 35.$$

## 8. More Practice

Example 4: Evaluate the expression

$$3x^2 - 4x + 8$$

when

$$x = -2.$$

### Solution

Substitute  $-2$  for  $x$ :

$$3(-2)^2 - 4(-2) + 8.$$

Evaluate the exponent:

$$3(4) - 4(-2) + 8.$$

Multiply:

$$12 + 8 + 8.$$

Add:

$$28.$$

Example 5: Simplify:

$$4(2x + 3) - 5(x - 1).$$

### Solution

Distribute:

$$4(2x + 3) - 5(x - 1) = 8x + 12 - 5x + 5.$$

Combine like terms:

$$(8x - 5x) + (12 + 5)$$

$$= 3x + 17.$$

Example 6: Simplify:

$$3(4x^2 - 2) - 5(7 - 6x^2).$$

### Solution

Distribute:

$$3(4x^2 - 2) - 5(7 - 6x^2) = 12x^2 - 6 - 35 + 30x^2.$$

Group like terms:

$$= (12x^2 + 30x^2) - 6 - 35.$$

Combine like terms:

$$= 42x^2 - 41.$$

## Summary

### Key Ideas

- Variables are letters used to represent numbers.
- An algebraic expression combines numbers, variables, and operation symbols.
- The order of operations is used to evaluate algebraic expressions.
- A formula is an equation that expresses a relationship between variables.
- Mathematical models use formulas to describe real-world situations.
- Terms are parts of an expression separated by addition or subtraction.
- A coefficient is the numerical part of a term.
- Like terms have the same variable factors.
- The distributive property helps simplify expressions.